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Studierichting: Informatica
Oefeningenreeks 1D nr. 9.2

$$\begin{aligned} & z^4 - 7z^3 + 7z^2 - 7z + 7 \\ \Rightarrow & (3 + \sqrt{7})^4 - 7(3 + \sqrt{7})^3 + 7(3 + \sqrt{7})^2 - 7(3 + \sqrt{7}) + 7 \\ = & (3 + \sqrt{7})^4 - 7(3 + \sqrt{7})^3 + 7 \cdot (9 + 2 \cdot 3\sqrt{7} + 7) - 21 - 7\sqrt{7} + 7 \\ = & (3 + \sqrt{7})^4 - 7(3 + \sqrt{7})^3 + 7[16 + 6\sqrt{7}] - 21 - 7\sqrt{7} + 7 \\ = & (16 + 6\sqrt{7})(16 + 6\sqrt{7}) - 7(16 + 6\sqrt{7})(3 + \sqrt{7}) + 112 + 42\sqrt{7} - 21 - 7\sqrt{7} + 7 \\ = & (16 + 6\sqrt{7})(16 + 6\sqrt{7}) - 7(16 + 6\sqrt{7})(3 + \sqrt{7}) + 98 + 35\sqrt{7} \\ = & 256 + 96\sqrt{7} + 96\sqrt{7} + 36 \cdot 7 + (-112 - 42\sqrt{7})(3 + \sqrt{7}) + 98 + 35\sqrt{7} \\ = & 508 + 192\sqrt{7} + (-336 - 112\sqrt{7} - 126\sqrt{7} - 42 \cdot 7) + 98 + 35\sqrt{7} \\ = & 508 + 192\sqrt{7} - 630 - 238\sqrt{7} + 98 + 35\sqrt{7} \\ = & -24 - 11\sqrt{7} \end{aligned}$$

References